

# Digital Manufacturing Value Case & KPI Impact Model

Directional executive value case for MES-enabled operations

- Value Case
- KPIs
- ROI
- Executive Summary

**Executive summary**

Connects manufacturing pain points to directional KPI movement without making unsupported ROI claims.

**Problem statement**

Digital manufacturing initiatives are often justified with broad efficiency language. Leaders need a clearer link between shop-floor friction, review effort, release readiness, data visibility, and measurable business value.

**Target-state recommendation**

Frame the value case around directional KPI movement: reduced manual effort, improved right-first-time execution, clearer release readiness, better deviation context, and more timely manufacturing visibility.

**Value drivers**

- Deviation effort and context gathering.
- Batch review time and release readiness.
- Right-first-time execution and documentation quality.
- Manufacturing visibility and leadership decision support.

**Assumptions**

- Values are illustrative and directional, not financial commitments.
- Site-specific baselines are required before business-case approval.
- Benefits depend on adoption, master data quality, and sustained process governance.

**Success criteria**

- KPIs are owned across operations, QA, and digital teams.
- Leadership can see how MES improvements connect to measurable value.
- Results are measured against agreed baseline data.

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## Risks, dependencies, and KPI impact model

**Risk and dependency framing**

- Directional value estimates require validation with actual site data.
- KPI ownership must be agreed across operations, QA, and digital teams.
- Benefits depend on adoption, master data quality, and sustained governance.

| Directional KPI / success measures |                                           |                                                  |
|------------------------------------|-------------------------------------------|--------------------------------------------------|
| Measure                            | Baseline / current signal                 | Target-state signal                              |
| Deviation effort                   | Manual context gathering                  | Richer triage context                            |
| Batch review time                  | Readiness discovered late                 | Earlier missing-item visibility                  |
| Right-first-time execution         | Corrections and rework difficult to trend | Structured checks and clearer exception patterns |
| Manufacturing visibility           | Status reporting delayed                  | Shared execution and release readiness view      |

Values and examples are illustrative and directional. Validate assumptions with site-specific data before decision-making.